

Códigos Utilizados no Databricks

- Leitura do Dataset

```
df_basics = spark.read.csv(  
    "dbfs:/Volumes/workspace/mvp/filmes_imdb/title.basics.tsv.gz",  
    sep="\t",  
    header=True,  
    inferSchema=True  
)
```

```
df_ratings = spark.read.csv(  
    "dbfs:/Volumes/workspace/mvp/filmes_imdb/title.ratings.tsv.gz",  
    sep="\t",  
    header=True,  
    inferSchema=True  
)
```

- Filtra filmes e colunas principais

```
from pyspark.sql.functions import col
```

```
df_filmes_tratado = (  
    df_basics  
    .filter(col("titleType") == "movie")  
    .select(  
        "tconst",  
        "primaryTitle",  
        "startYear",
```

```
        "runtimeMinutes",  
        "genres"  
    )  
)
```

- Print schema

```
df_basics.printSchema()
```

- Filtrando nulos

```
#Criando Views
```

```
df_filmes_tratado.createOrReplaceTempView("filmes")
```

```
df_ratings.createOrReplaceTempView("ratings")
```

```
-- Verificando Nulos - Tabela Filmes
```

```
SELECT
```

```
    COUNT(*) AS total_registros,
```

```
    SUM(CASE WHEN startYear IS NULL THEN 1 ELSE 0 END) AS  
startYear_nulos,
```

```
    SUM(CASE WHEN runtimeMinutes IS NULL THEN 1 ELSE 0 END) AS  
runtime_nulos,
```

```
    SUM(CASE WHEN genres IS NULL THEN 1 ELSE 0 END) AS genres_nulos
```

```
FROM filmes;
```

```
-- Verificando Nulos - Tabela Rating
```

```
SELECT
```

```
    COUNT(*) AS total_registros,
```

```
SUM(CASE WHEN averageRating IS NULL THEN 1 ELSE 0 END) AS  
rating_nulos,  
SUM(CASE WHEN numVotes IS NULL THEN 1 ELSE 0 END) AS votos_nulos  
FROM ratings;
```

- Quantidade de filmes

--Verificando a quantidade de filmes

```
SELECT COUNT(*) AS total_filmes  
FROM filmes;
```

- Quais gêneros têm maior média de avaliação no IMDb?

```
SELECT  
f.genres,  
AVG(r.averageRating) AS media_avaliacao,  
COUNT(*) AS quantidade_filmes  
FROM filmes f  
JOIN ratings r  
ON f.tconst = r.tconst  
GROUP BY f.genres  
ORDER BY media_avaliacao DESC;
```

- Existe uma relação entre o ano de lançamento e a avaliação do filme?

```
SELECT startYear, AVG(averageRating)  
FROM filmes f  
JOIN ratings r ON f.tconst = r.tconst
```

GROUP BY startYear

ORDER BY startYear;

--Relação do ano de lançamento com avaliação por década

SELECT

FLOOR(CAST(f.startYear AS INT) / 10) * 10 AS decada,

ROUND(AVG(r.averageRating), 2) AS media_avaliacao,

COUNT(*) AS quantidade_filmes

FROM filmes f

JOIN ratings r

ON f.tconst = r.tconst

WHERE f.startYear IS NOT NULL

AND f.startYear <> '\N'

GROUP BY FLOOR(CAST(f.startYear AS INT) / 10) * 10

ORDER BY decada;

- A duração do filme impacta sua avaliação?

SELECT

CASE

WHEN try_cast(f.runtimeMinutes AS INT) < 60 THEN 'Até 60 min'

WHEN try_cast(f.runtimeMinutes AS INT) BETWEEN 60 AND 89 THEN '60–89 min'

WHEN try_cast(f.runtimeMinutes AS INT) BETWEEN 90 AND 119 THEN '90–119 min'

WHEN try_cast(f.runtimeMinutes AS INT) BETWEEN 120 AND 149 THEN '120–149 min'

ELSE '150 min ou mais'

END AS faixa_duracao,

```

ROUND(AVG(r.averageRating), 2) AS media_avaliacao,
COUNT(*) AS quantidade_filmes
FROM filmes f
JOIN ratings r
ON f.tconst = r.tconst
WHERE try_cast(f.runtimeMinutes AS INT) IS NOT NULL
GROUP BY
CASE
    WHEN try_cast(f.runtimeMinutes AS INT) < 60 THEN 'Até 60 min'
    WHEN try_cast(f.runtimeMinutes AS INT) BETWEEN 60 AND 89 THEN '60–
89 min'
    WHEN try_cast(f.runtimeMinutes AS INT) BETWEEN 90 AND 119 THEN '90–
119 min'
    WHEN try_cast(f.runtimeMinutes AS INT) BETWEEN 120 AND 149 THEN
'120–149 min'
    ELSE '150 min ou mais'
END
ORDER BY faixa_duracao;

```

```

SELECT
CASE
    WHEN r.numVotes < 100 THEN 'Até 100 votos'
    WHEN r.numVotes BETWEEN 100 AND 999 THEN '100–999 votos'
    WHEN r.numVotes BETWEEN 1000 AND 9999 THEN '1k–9k votos'
    WHEN r.numVotes BETWEEN 10000 AND 99999 THEN '10k–99k votos'
    ELSE '100k votos ou mais'
END AS faixa_votos,
ROUND(AVG(r.averageRating), 2) AS media_avaliacao,
COUNT(*) AS quantidade_filmes

```

```

FROM ratings r
JOIN filmes f
  ON r.tconst = f.tconst
WHERE r.numVotes IS NOT NULL
GROUP BY
  CASE
    WHEN r.numVotes < 100 THEN 'Até 100 votos'
    WHEN r.numVotes BETWEEN 100 AND 999 THEN '100–999 votos'
    WHEN r.numVotes BETWEEN 1000 AND 9999 THEN '1k–9k votos'
    WHEN r.numVotes BETWEEN 10000 AND 99999 THEN '10k–99k votos'
    ELSE '100k votos ou mais'
  END
ORDER BY faixa_votos;

```

- Filmes com mais votos têm melhores notas?

```

SELECT
  CASE
    WHEN r.numVotes < 100 THEN 'Até 100 votos'
    WHEN r.numVotes BETWEEN 100 AND 999 THEN '100–999 votos'
    WHEN r.numVotes BETWEEN 1000 AND 9999 THEN '1k–9k votos'
    WHEN r.numVotes BETWEEN 10000 AND 99999 THEN '10k–99k votos'
    ELSE '100k votos ou mais'
  END AS faixa_votos,
  ROUND(AVG(r.averageRating), 2) AS media_avaliacao,
  COUNT(*) AS quantidade_filmes
FROM ratings r
JOIN filmes f
  ON r.tconst = f.tconst

```

WHERE r.numVotes IS NOT NULL

GROUP BY

CASE

WHEN r.numVotes < 100 THEN 'Até 100 votos'

WHEN r.numVotes BETWEEN 100 AND 999 THEN '100–999 votos'

WHEN r.numVotes BETWEEN 1000 AND 9999 THEN '1k–9k votos'

WHEN r.numVotes BETWEEN 10000 AND 99999 THEN '10k–99k votos'

ELSE '100k votos ou mais'

END

ORDER BY faixa_votos;

- Quais gêneros concentram os filmes mais populares (com maior número de votos)?

SELECT

f.genres,

SUM(r.numVotes) AS total_votos,

COUNT(*) AS quantidade_filmes

FROM filmes f

JOIN ratings r

ON f.tconst = r.tconst

WHERE f.genres IS NOT NULL

GROUP BY f.genres

ORDER BY total_votos DESC

LIMIT 10;